THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 5: **Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a circle of radius 13 cm, a chord is 5 cm from the centre. The length of the chord is: 1

(a) 26 cm **(b)** 12 cm **(c)** 24 cm **(d)** 22 cm

2. In a circle of radius 25 cm, a chord is 7 cm from the centre. The length of the chord is: 1

(a) 46 cm **(b)** 24 cm **(c)** 50 cm **(d)** 48 cm

3. A chord of length 16 cm is at a distance of 6 cm from the centre of a circle. The radius of the circle is: 1

(a) 10 cm **(b)** 16 cm **(c)** 14 cm **(d)** 11 cm

4. A chord of length 18 cm is at a distance of 12 cm from the centre of a circle. The radius of the circle is: 1

(a) 18 cm **(b)** 16 cm **(c)** 15 cm **(d)** 21 cm

5. A chord of length 12 cm is drawn in a circle of radius 10 cm. The distance of the chord from the centre is: 1

(a) 8 cm **(b)** 16 cm **(c)** 7 cm **(d)** 4 cm

6. An arc subtends an angle of 130° at the centre. The angle it subtends at a point on the remaining part of the circle is: 1

(a) 130° **(b)** 260° **(c)** 65° **(d)** 115°

7. ABCD is a cyclic quadrilateral with $\angle A = 110^\circ$. Then $\angle C$ is: 1

(a) 110° **(b)** 250° **(c)** 70° **(d)** 35°

8. An arc subtends 28° at a point on the circle. The angle subtended by the same arc at the centre is: 1

(a) 152° **(b)** 28° **(c)** 14° **(d)** 56°

9. Equal chords of a circle are at _____ distances from the centre. 1

(a) equal **(b)** double **(c)** unequal **(d)** half

10. Points lying on the same circle are called: 1

(a) concyclic points **(b)** coplanar points

(c) collinear points **(d)** concurrent points

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A chord of a circle of radius 17 cm is 8 cm away from the centre. Find its length. 2

12. A chord of length 40 cm is 21 cm from the centre of a circle. Find the radius. 2

13. A cyclic quadrilateral has angles $\angle A = 80°, \angle B = 110°, \angle C = 100°, \angle D = 70°$. Can it be drawn? Why? 2

14. An arc subtends $70°$ at the centre. Find the angle it subtends at a point on the circle outside the arc. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the perpendicular from the centre of a circle to a chord bisects the chord. 3

16. Prove that the angle subtended by a diameter at any point on the circle is $90°$. 3

17. Prove that the sum of the opposite angles of a cyclic quadrilateral ABCD is $180°$ (use the angle-at-centre theorem). 3

OR

A chord is 9 cm from the centre of a circle of radius 15 cm. Find its length. 3

18. Two chords AB (30 cm) and CD (16 cm) lie on the same side of the centre of a circle of radius 17 cm. Find the distance between them. 3

OR

Two parallel chords of length 6 cm and 8 cm are in a circle of radius 5 cm. Find the distance between them if they are on opposite sides of the centre. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the angle subtended by an arc at the centre of a circle is double the angle subtended by it at any point on the circle outside the arc (take the case where DC produced meets the circle at E on the arc). 5

OR

Prove that the sum of the opposite angles of a cyclic quadrilateral is $180°$. 5

20. Prove that of two chords of a circle, the longer chord is nearer to the centre. 5

OR

Case study: A circular park has radius 25 m. Two straight paths (chords) of lengths 14 m and 48 m are laid. 5

(i) Find the distance of each path from the centre.

(ii) Which is nearer the centre and why?

(iii) Find the distance between the paths if they are parallel and on opposite sides of the centre.

(iv) Find it if they are on the same side.

END OF QUESTION PAPER — Total: 40 Marks